

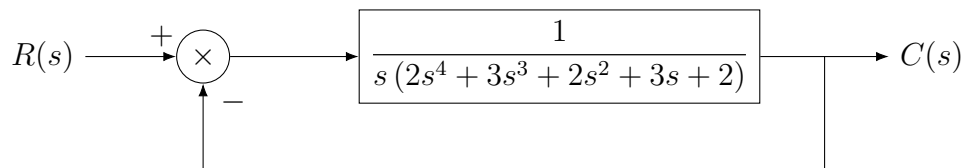
- Q1.** Construct the Routh table and determine how many roots of the following polynomial lie in the right half-plane (RHP) and in the left half-plane (LHP):

$$P(s) = 3s^7 + 9s^6 + 6s^5 + 4s^4 + 7s^3 + 8s^2 + 2s + 6.$$

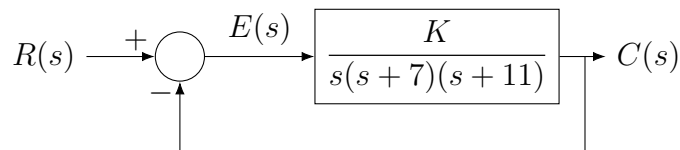
- Q2.** Use the Routh–Hurwitz criterion to determine how many poles of the following closed-loop system, $T(s)$, lie in the RHP, in the LHP, and on the $j\omega$ -axis:

$$T(s) = \frac{s^3 + 7s^2 - 21s + 10}{s^6 + s^5 - 6s^4 + 0s^3 - s^2 - s + 6}.$$

- Q3.** Find the number of poles in the LHP, the RHP, and on the $j\omega$ -axis for the system.



- Q4.** Find the range of gain, K , for the system that will cause the system to be stable, unstable, and marginally stable. Assume $K > 0$.



- Q5.** Given the state-space system

$$\dot{x} = \begin{bmatrix} 0 & 3 & 1 \\ 2 & 8 & 1 \\ -10 & -5 & -2 \end{bmatrix} x + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u,$$

with output equation

$$y = [1 \ 0 \ 0] x,$$

determine how many poles lie in the left half-plane, in the right half-plane, and on the $j\omega$ -axis.

- Q6.** For the system shown in Figure, determine the value of gain K that will cause the system to oscillate. Also, determine the frequency of oscillation.

